Why Visualize? The Case for Data Viz

Workshop 1 — Module 1

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2025/26

Data Visualization for Data Scientists
UNYT Tirana

Module Roadmap

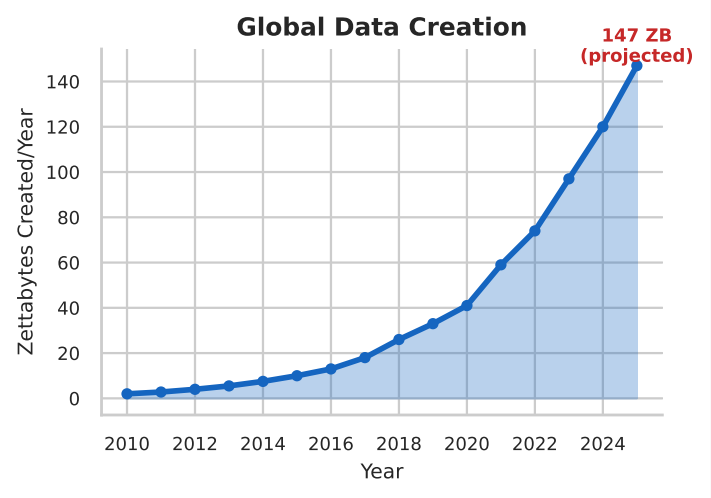
1. The information overload problem
2. Historical masterpieces of data visualization
3. Anscombe's quartet: why summary statistics lie
4. The Datasaurus Dozen: a modern extension
5. What makes visualization "good"?

Learning Outcome
After this module you will be able to articulate *why* visualization is an indispensable analytical tool, not merely a decorative afterthought.

The Information Overload Problem

We Are Drowning in Data

- 2.5 quintillion bytes of data created *daily*
- 90% of the world's data generated in the last two years
- Corporate databases double every 12–18 months
- Human working memory: 7 ± 2 chunks



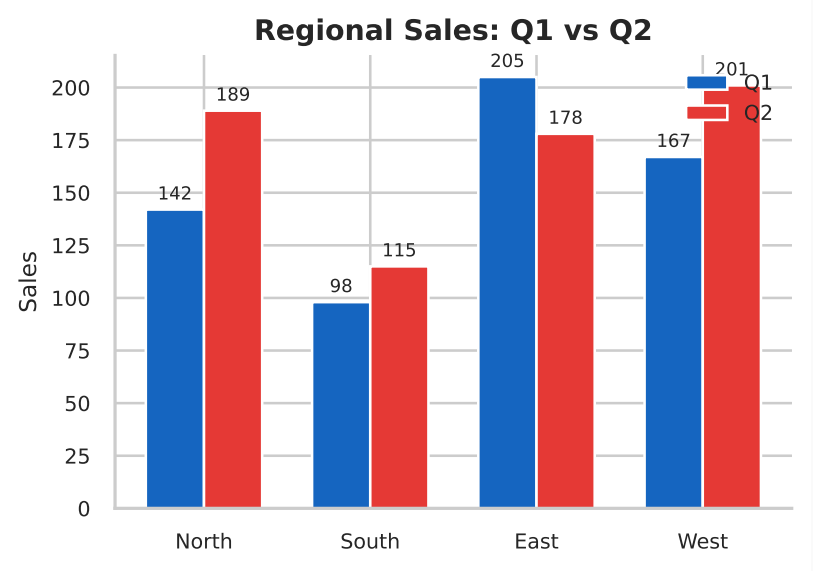
Pro-Tip
Raw tables with more than ∼20 rows are cognitively impenetrable. Visualization compresses dimensionality for the human visual cortex.

The Visual Bandwidth Advantage

Table (10 seconds to parse):

Region	Q1	Q2
North	142	189
South	98	115
East	205	178
West	167	201

Chart (<1 second to grasp):



Data Insight
The visual cortex processes spatial patterns ~60,000× faster than sequential text. Charts exploit this parallel processing.

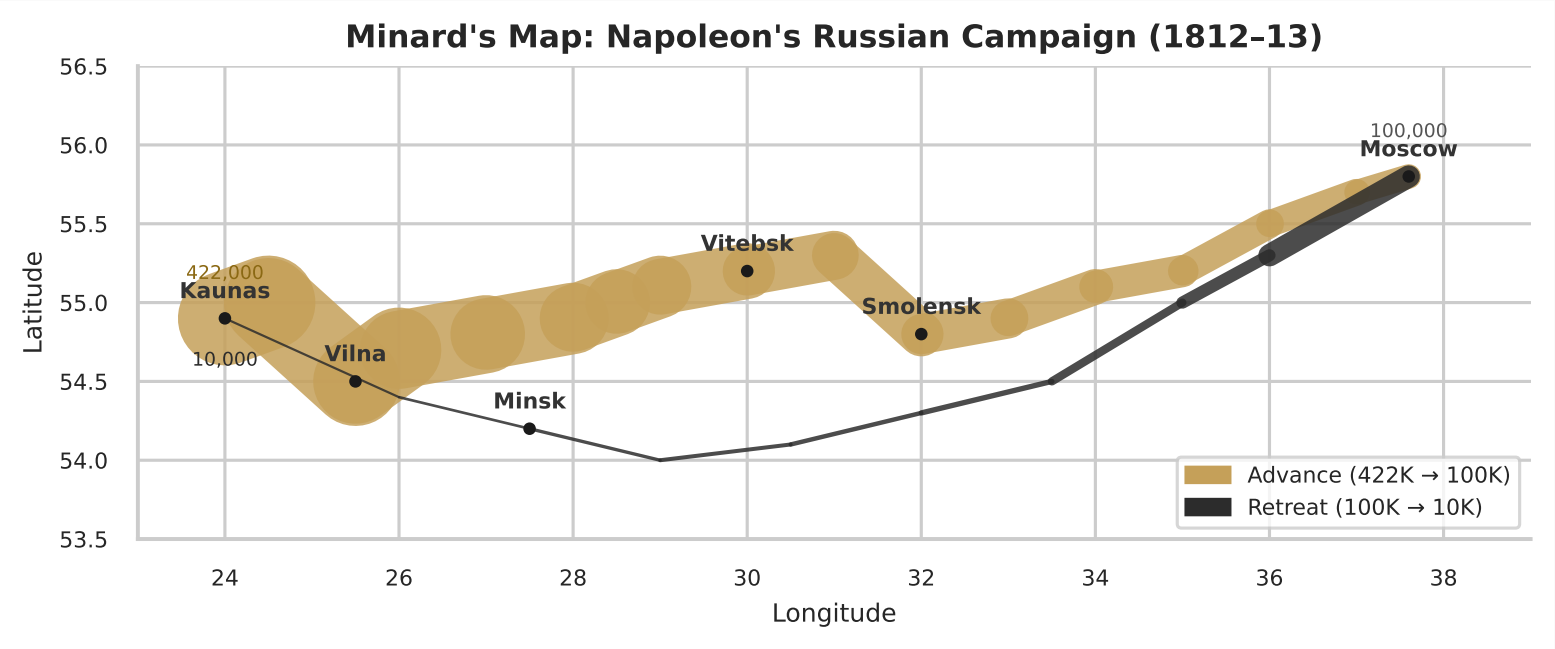
Three Purposes of Visualization

1. **Exploratory** — discover patterns, generate hypotheses
   "What is going on in my data?"
2. **Explanatory** — communicate known findings
   "Here is what I found and why it matters."
3. **Confirmatory** — validate or falsify expectations
   "Does the data support my model?"

Pro-Tip
Most analysts jump straight to explanatory charts. Always begin with exploratory visualization — you cannot explain what you have not first discovered.

Historical Masterpieces

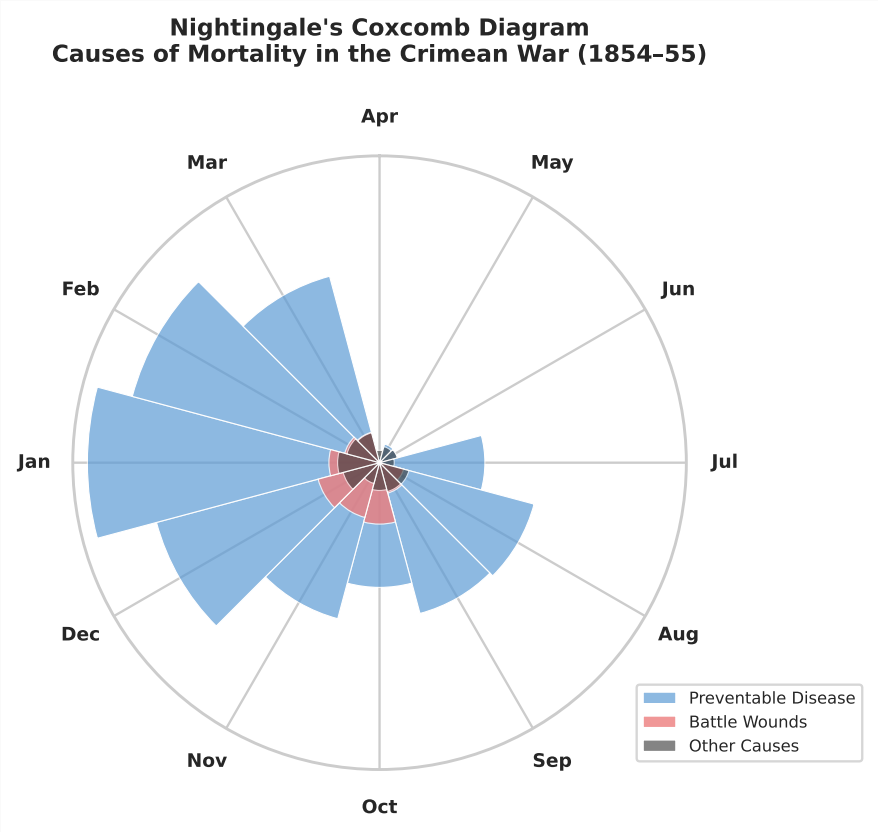
Charles Joseph Minard — Napoleon's March (1869)



Data Insight

Six variables on a single 2-D plane: army size, geographic position (lat/lon), direction of movement, temperature, and date. Tufte calls it "the best statistical graphic ever drawn."

Florence Nightingale — Coxcomb Diagram (1858)



- Showed preventable deaths in the Crimean War
- Persuaded Parliament to fund sanitary reforms
- Area encoding for monthly mortality counts
- A pioneer of *persuasive* data visualization

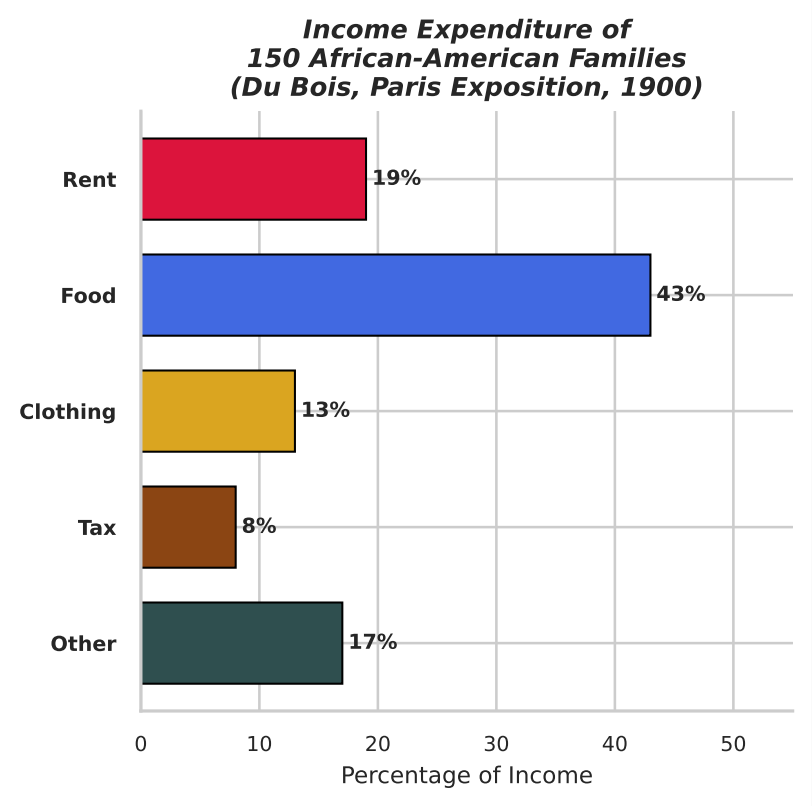
John Snow — Cholera Map (1854)

- Plotted cholera deaths as stacked bars on a street map
- Spatial clustering around the Broad Street pump
- Convinced authorities to remove the pump handle
- Foundational example of spatial epidemiology

Data Insight

Snow's map is arguably the first GIS analysis—a point pattern overlaid on a geographic reference layer, centuries before ArcGIS.

W.E.B. Du Bois — Data Portraits (1900)



- Presented at the Paris Exposition, 1900
- Visualized African-American economic progress
- Strikingly modern design: spiral charts, proportional areas
- Data as a tool of social justice and advocacy

Lessons from the Masters

Practitioner	Year	Key Innovation
Minard	1869	Multivariate narrative on a single plane
Nightingale	1858	Persuasive polar-area chart
Snow	1854	Spatial point-pattern overlay
Du Bois	1900	Design as social advocacy

Every masterpiece combined **data integrity** with **design clarity** and a **compelling narrative**.

Anscombe's Quartet

The Trap of Summary Statistics

Four datasets, *identical* descriptive statistics:

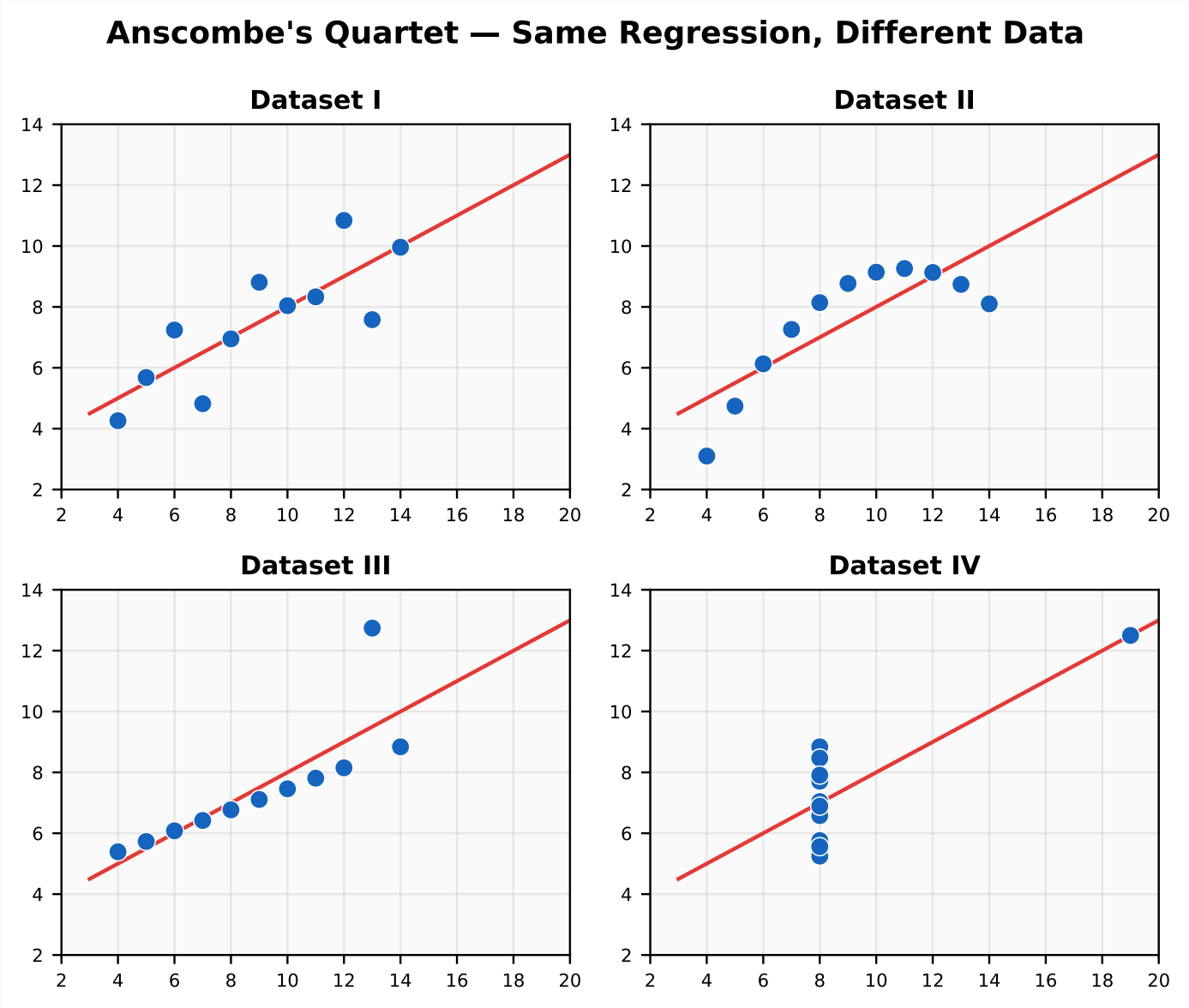
- $\bar{x} = 9.0, \quad \bar{y} = 7.50$
- $\sigma_x = 3.32, \quad \sigma_y = 2.03$
- $r = 0.816$
- Regression: $y = 3.0 + 0.5x$

Anscombe's Quartet: Identical Summary Statistics

Stat	Dataset I	Dataset II	Dataset III	Dataset IV
Mean x	9.0	9.0	9.0	9.0
Mean y	7.50	7.50	7.50	7.50
SD x	3.32	3.32	3.32	3.32
SD y	2.03	2.03	2.03	2.03
r	0.816	0.816	0.816	0.817

Question: If every statistical summary is identical, are the datasets interchangeable?

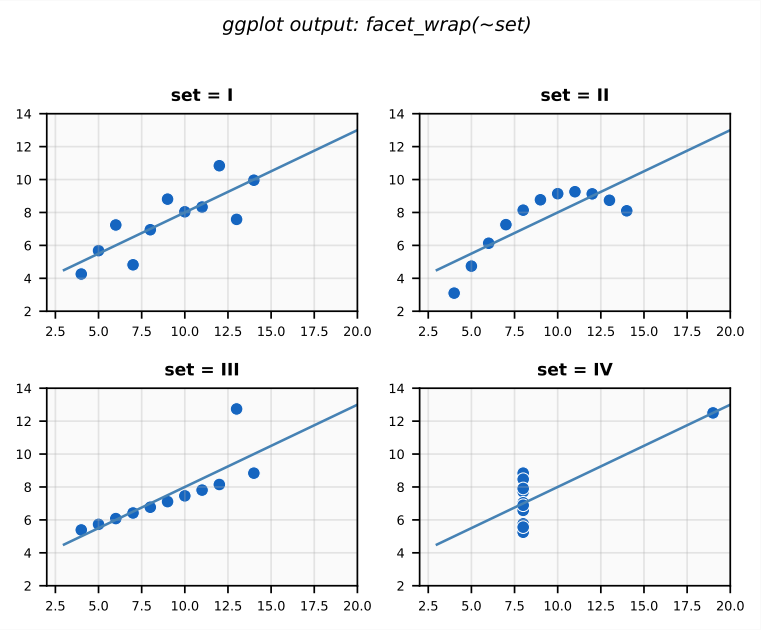
Anscombe's Quartet — Visualized



Data Insight

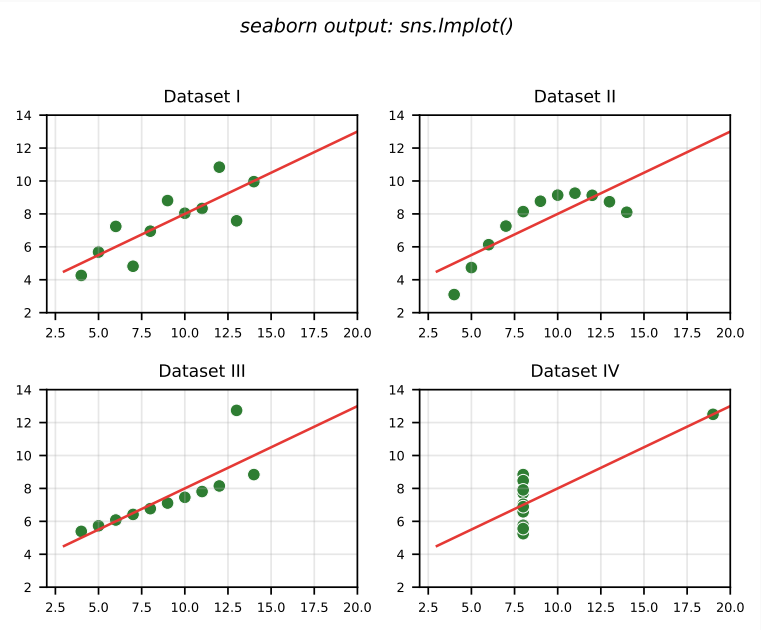
Anscombe's Quartet in R

```r
# Anscombe's quartet is built into
    R
data(anscombe)

# Reshape to long format
library(tidyverse)
anscombe_long <- anscombe |>
  pivot_longer(
    everything(),
    names_to = c(".value", "set"),
    names_pattern = "(.)(.)"
  )

# Plot all four datasets
ggplot(anscombe_long,
       aes(x = x, y = y)) +
  geom_point(size = 2) +
  geom_smooth(method = "lm",
              se = FALSE,
              color = "steelblue")
                  +
  facet_wrap(~set, ncol = 2) +
  theme_minimal()
```



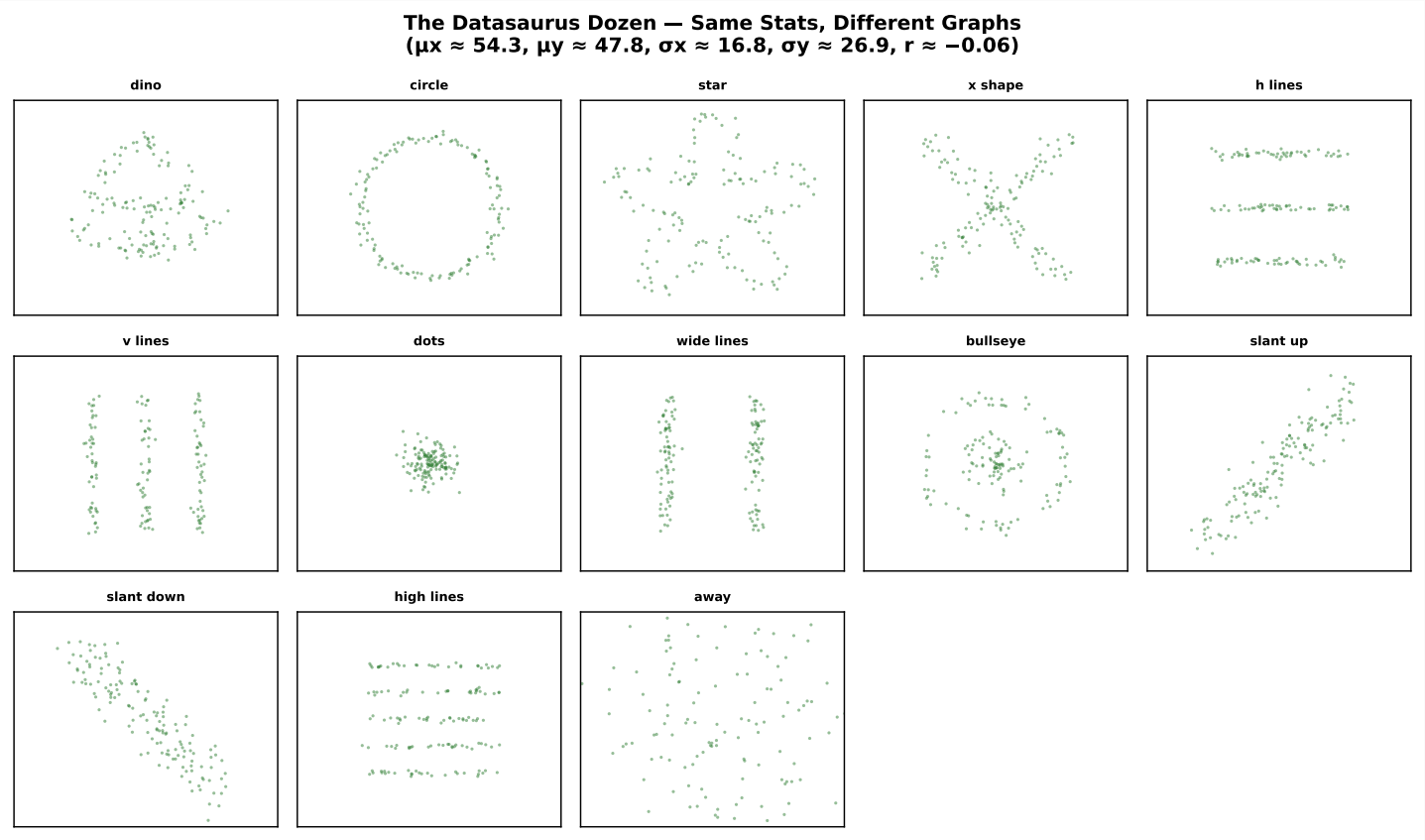
Anscombe's Quartet in Python

```python
import seaborn as sns
import matplotlib.pyplot as plt

# Load built-in Anscombe data
df = sns.load_dataset("anscombe")

# FacetGrid with regression
g = sns.lmplot(
    data=df,
    x="x", y="y",
    col="dataset",
    col_wrap=2,
    ci=None,
    height=3,
    scatter_kws={"s": 40}
)
g.set_titles("Dataset {col_name}")
plt.tight_layout()
plt.savefig("anscombe.png", dpi
    =150)
```



The Datasaurus Dozen

Beyond Anscombe: The Datasaurus Dozen (2017)

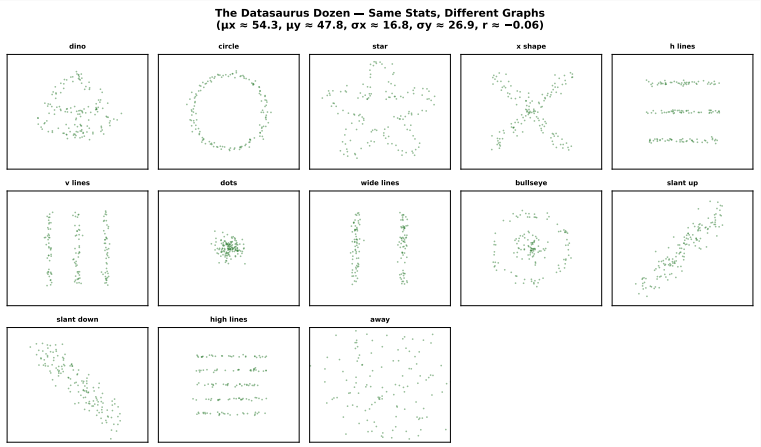


Data Insight

Matejka & Fitzmaurice (2017) generated 13 datasets with *identical* means, standard deviations, and correlations — yet wildly different shapes, including a dinosaur. Reference: *CHI 2017 Proceedings*.

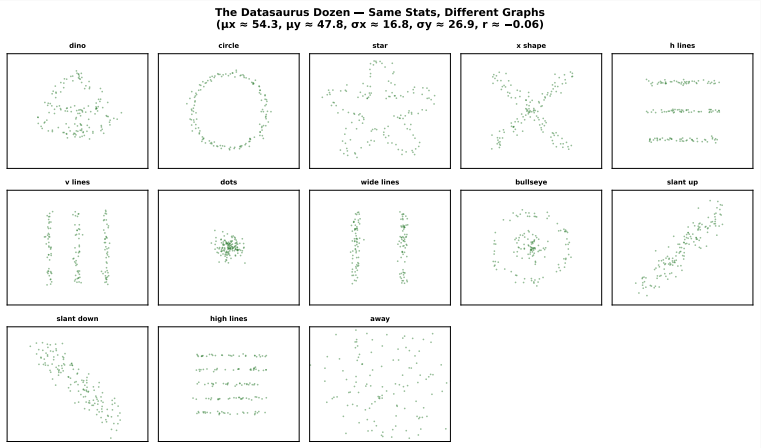
Datasaurus in R

```r
# Install datasauRus package
# install.packages("datasauRus")
library(datasauRus)
library(ggplot2)

# All 13 datasets share the
# same summary statistics
ggplot(datasaurus_dozen,
       aes(x = x, y = y)) +
  geom_point(alpha = 0.6,
             size = 0.8) +
  facet_wrap(~dataset,
             ncol = 5) +
  theme_minimal(base_size = 8) +
  labs(
    title = "Same Stats,
             Different Graphs",
    caption = "datasauRus package"
  )
```



Datasaurus in Python

```python
# pip install datasaurus
# or use the CSV directly
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

url = ("https://raw.
     githubusercontent"
        ".com/jumpingrivers/"
        "datasauRus/main/inst/"
        "extdata/DatasaurusDozen"
        "-long.tsv")
df = pd.read_csv(url, sep="\t")

g = sns.FacetGrid(
    df, col="dataset",
    col_wrap=5, height=2
)
g.map(plt.scatter, "x", "y",
      s=5, alpha=0.6)
g.set_titles("{col_name}")
plt.tight_layout()
plt.savefig("datasaurus.png",
            dpi=150)
```



Verify: Same Statistics, Different Pictures

R verification:

```r
library(dplyr)
datasaurus_dozen |>
  group_by(dataset) |>
  summarise(
    mean_x = mean(x),
    mean_y = mean(y),
    sd_x   = sd(x),
    sd_y   = sd(y),
    cor_xy = cor(x, y)
  )
```

Python verification:

```python
summary = df.groupby(
    "dataset"
).agg(
    mean_x=("x", "mean"),
    mean_y=("y", "mean"),
    sd_x=("x", "std"),
    sd_y=("y", "std"),
    cor_xy=("x", lambda s:
      s.corr(
        df.loc[s.index, "y"]
      ))
)
print(summary.round(2))
```

Pro-Tip

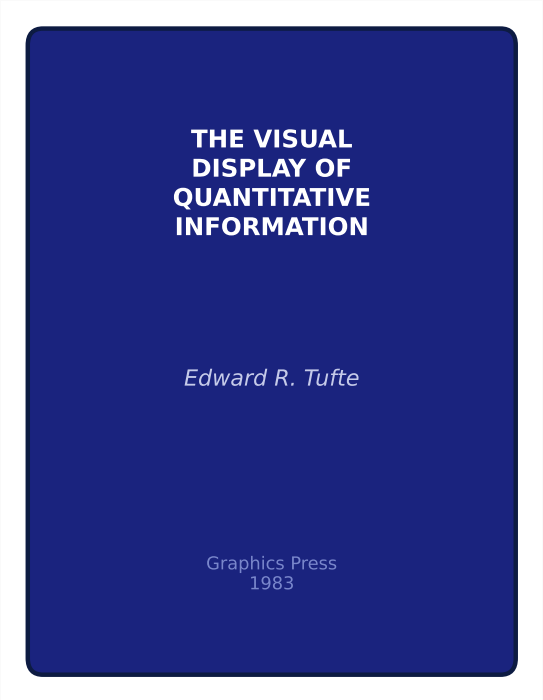
Every dataset returns $\bar{x} \approx 54.3$, $\bar{y} \approx 47.8$, $\sigma_x \approx 16.8$, $\sigma_y \approx 26.9$, $r \approx -0.06$. **Never trust statistics without visualization.**

What Makes Visualization "Good"?

Edward Tufte's Principles of Graphical Excellence

A visualization achieves **graphical excellence** when it:

1. Shows the data clearly and accurately
2. Induces thinking about *substance*, not method
3. Avoids distortion of what the data reveal
4. Presents many numbers in a small space



"Excellence in statistical graphics consists of complex ideas communicated with clarity, precision, and efficiency."

— Tufte (1983), p. 51

Alberto Cairo's Truthful Art Framework

A visualization should be:

1. **Truthful** — based on thorough, honest research
2. **Functional** — constitutes an accurate depiction
3. **Beautiful** — attractive and intriguing
4. **Insightful** — reveals evidence not detectable otherwise
5. **Enlightening** — changes minds if warranted by evidence

Data Insight
Cairo's framework resolves the false dichotomy between "pretty" and "accurate." Effective visualization is *simultaneously* both.

The Data-Ink Ratio

$$\text{Data-Ink Ratio} = \frac{\text{Ink used to represent data}}{\text{Total ink on the graphic}}$$

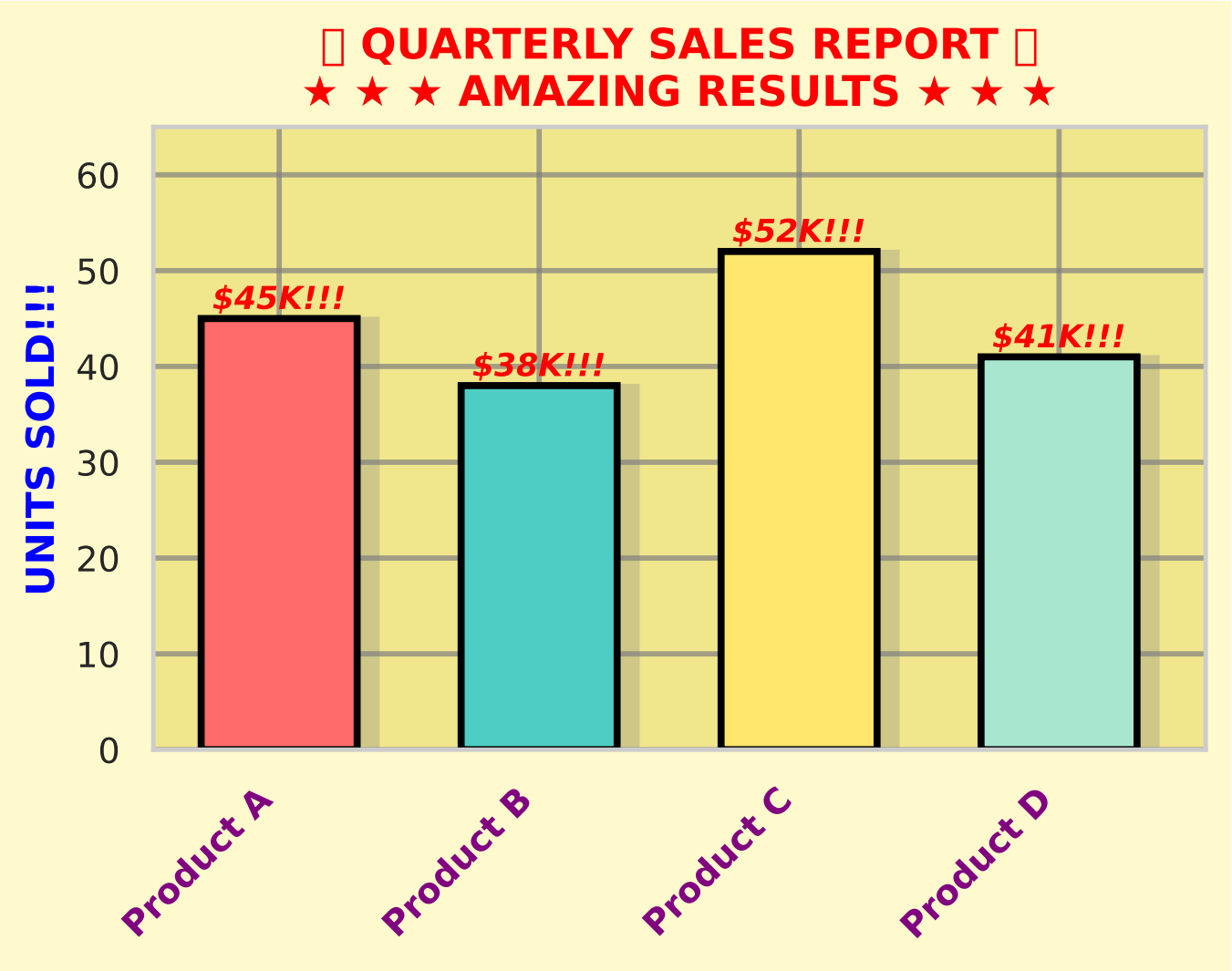
Low ratio (bad):

- Heavy gridlines
- 3-D effects, drop shadows
- Decorative images
- Redundant legends

High ratio (good):

- Minimal non-data ink
- Direct labels
- Whitespace as structure
- Every mark = a data point

Chart Junk: What to Avoid

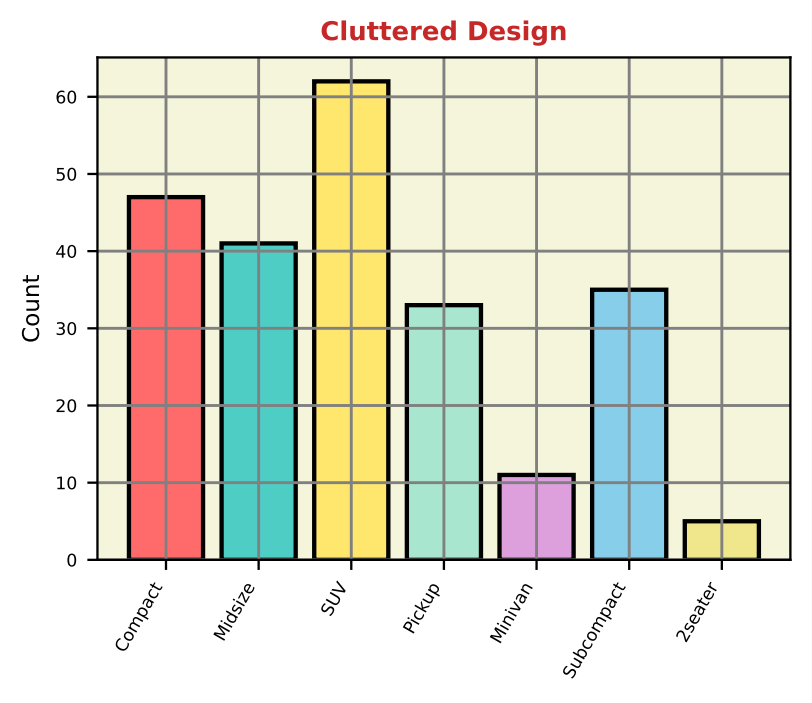


Tufte's catalogue of chart junk:

- Unnecessary 3-D effects and perspective distortion

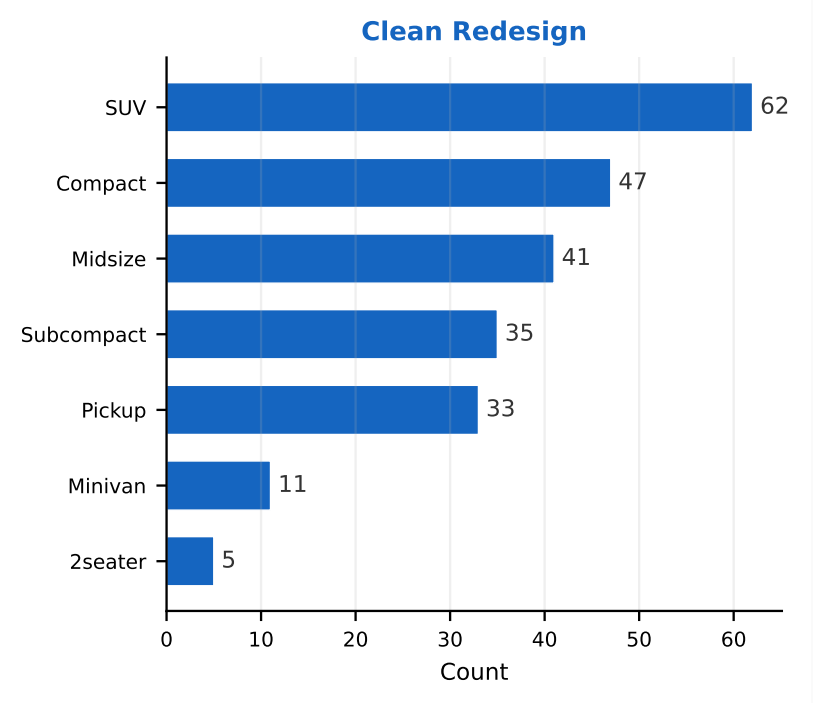
Good vs. Bad: A Before/After

Before (Cluttered)



Issues: 3-D bars, gradient fills, rotated axis labels, unnecessary legend box.

After (Clean)



Fixes: 2-D horizontal bars, direct labels, sorted by value, minimal grid.

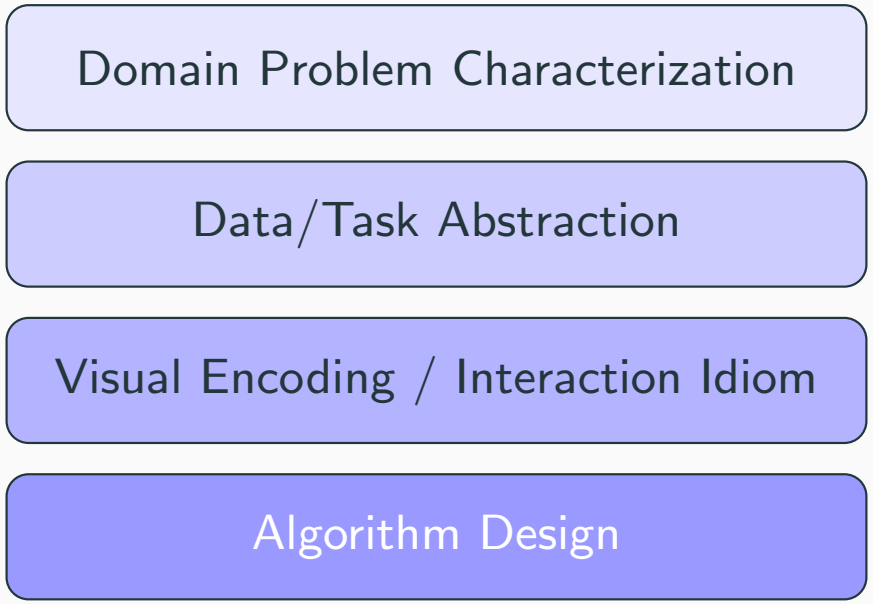
The Lie Factor

$$\text{Lie Factor} = \frac{\text{Effect shown in graphic}}{\text{Effect in data}}$$

- A lie factor of 1.0 = perfect honesty
- Lie factor > 1.05 or < 0.95 = visual deception
- Common sources: truncated axes, area-encoded bars, perspective

Pro-Tip
Always check: if the data increased by 50%, does the visual encoding (height, area, volume) also increase by exactly 50%?

Munzner's Nested Model for Visualization Design



Each layer validates the layer above it. Errors at outer layers propagate inward and cannot be fixed by inner-layer improvements.

Putting It Together

The Visualization Design Checklist

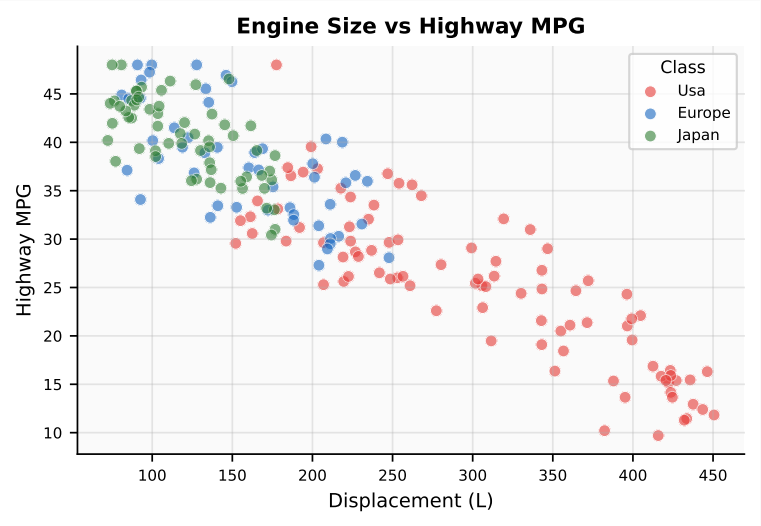
Before publishing any chart, verify:

1. **Purpose**: What question does this answer?
2. **Data integrity**: Does it accurately represent the data?
3. **Audience**: Who will read this and what do they need?
4. **Clarity**: Can the message be grasped in < 5 seconds?

Data Insight
Effective visualization sits at the intersection of **data science** (what is true), **design** (what is clear), and **storytelling** (what is compelling).

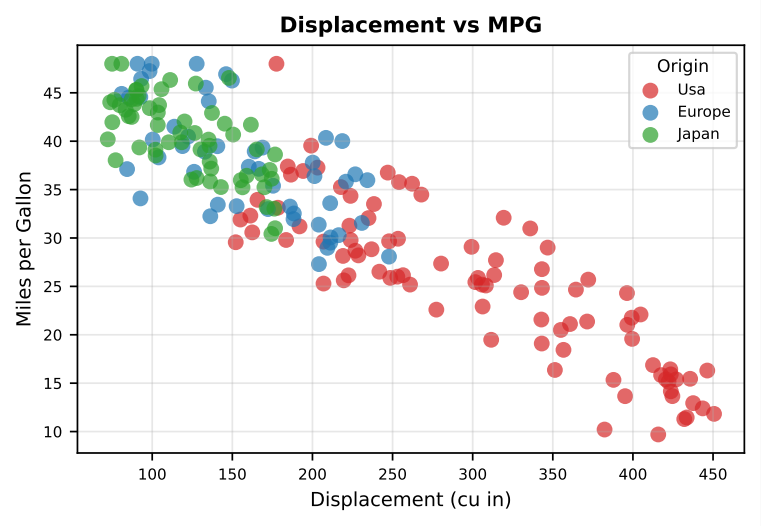
Your First Plot in R (Preview)

```r
library(ggplot2)

# Built-in mpg dataset
ggplot(mpg,
       aes(x = displ,
           y = hwy,
           color = class)) +
  geom_point(size = 2,
             alpha = 0.7) +
  labs(
    title = "Engine Size vs
             Highway MPG",
    x = "Displacement (L)",
    y = "Highway MPG"
  ) +
  theme_minimal()
```



Your First Plot in Python (Preview)

```python
import seaborn as sns
import matplotlib.pyplot as plt

# Built-in mpg dataset
mpg = sns.load_dataset("mpg")

sns.scatterplot(
    data=mpg,
    x="displacement",
    y="mpg",
    hue="origin",
    alpha=0.7,
    s=50
)
plt.title("Displacement vs MPG")
plt.xlabel("Displacement (cu in)")
plt.ylabel("Miles per Gallon")
plt.tight_layout()
plt.savefig("first_plot.png",
            dpi=150)
```



Key Takeaways

1. **Visualization is analysis**, not decoration — it reveals structure that summary statistics hide (Anscombe, Datasaurus).
2. **Historical precedent**: Minard, Nightingale, Snow, Du Bois used visualization to change policy and save lives.
3. **Principled design** matters: data-ink ratio, lie factor, and the nested model provide rigorous evaluation criteria.
4. **R and Python** are equally capable — this course teaches both in parallel so you choose the right tool for the context.

Essential Readings for This Module

- Tufte, E. R. (1983). *The Visual Display of Quantitative Information*, Chapter 1.
- Cairo, A. (2012). *The Functional Art*, Introduction & Chapter 1.
- Anscombe, F. J. (1973). "Graphs in Statistical Analysis," *The American Statistician*, 27(1), 17–21.
- Matejka, J. & Fitzmaurice, G. (2017). "Same Stats, Different Graphs," *CHI 2017*.

Next Module: Tufte's Principles of Graphical Excellence (W01-M02)